O-C-828C
September 11, 1980
SUPERSEDING
Fed. Spec. O-C-828b
May 9, 1968

FEDERAL SPECIFICATION

CUPRIC SULFATE, PENTAHYDRATE, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers technical grade cupric sulfate, pentahydrate.

2. APPLICABLE DOCUMENTS

2.1 The following docments, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 536/2101 - Soap and Soap-Products (Including Synthetic Detergents): Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-B-117 - Bags, Sleeve and Tubing-Interior Packaging.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" by 48" Pallets.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Code of Federal Regulations (CFR):

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations.

(The Code of Federal Regulations and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standards:

C136 - Sieve or Screen Analysis of Fine and Coarse Aggregates.
D1193 - Reagent Water.
E11 - Wire-Cloth Sieves for Testing Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. Cupric sulfate, pentahydrate shall be blue, crystalline, and free from hard caking when tested as specified in 4.2.4.6.

3.2 Physical and chemical characteristics. Cupric sulfate, pentahydrate shall conform to the physical and chemical characteristics of table I when tested as specified therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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TABLE I. Physical and chemical characteristics

Characteristic	Percent by weight: Minimum	Maximum	Test paragraph
Assay (as $CuSO_4 \cdot 5H_2O$)	98.0	105.0	4.2.4.2
Insoluble matter	----	0.1	4.2.4.3
Iron (Fe)	----	0.2	4.2.4.4
Material passing a 3/4-inch (19 mm) sieve	100	----	4.2.4.5
Material retained on a 2.38 mm (No. 8) sieve	95	----	4.2.4.5

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the cupric sulfate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative sample of approximately 450 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for cupric sulfate test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Liner bag missing or not as specified (when required)
(f) Fiberboard packing aids missing or not as specified (where required)
(g) Marking incorrect, missing, or illegible
(h) Unitization not as specified

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 450 g. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. When applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Preparation of specimen. Crush rapidly without grinding about 100 g of the specimen until it will pass through a No. 16 sieve conforming to ASTM E11. This material shall be used for 4.2.4.2, 4.2.4.3 and 4.2.4.4. The specimen shall be tested as-received for 4.2.4.5 and 4.2.4.6.

4.2.4.2 Assay (as $CuSO_4 \cdot 5H_2O$).

(a) Starch indicator solution. Mix about 2 g of soluble starch with sufficient cold water to form a thin paste. Pour this paste into about 1 liter of boiling water, add 1 g of salicylic acid, and boil until the solution is clear. Cool and transfer to a glass-stoppered bottle.

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(b) Procedure. Weigh to nearest milligram approximately 20 g of the crushed specimen (see 4.2.4.1) and dissolve in water to make 1 liter in a volumetric flask. To 50 milliliters (ml) of the solution, measured with a pipet, add 4 ml of glacial acetic acid and cool to 8° ± 2°C. Add approximately 3 g of potassium iodide, dissolved in 5 ml of water, shake until dissolved, and titrate immediately with 0.1N sodium thiosulfate solution. When the solution becomes pale yellow, add 3 ml of starch indicator solution prepared as specified in (a), which changes the color to blue, and continue the titration until the color becomes yellowish-white. Calculate the percent cupric sulfate, pentahydrate as follows:

$$\text{Percent cupric sulfate, pentahydrate} = \frac{499.4AB}{W}$$

where: A = Milliliters of sodium thiosulfate solution used,
B = Normality of sodium thiosulfate solution, and
C = Weight of specimen in grams.

4.2.4.3 Insoluble matter. Weigh to the nearest milligram approximately 10 g of the crushed specimen (see 4.2.4.1), dissolve in 100 ml of water, and add 3 ml of sulfuric acid. Stir and allow to stand for 10 minutes at room temperature. Stir again and then filter through a tared filter crucible. Wash with approximately 50 ml of water, dry to constant weight at 105° to 110°C, cool in a desiccator, and weigh. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A - B)}{W}$$

where: A = Weight of crucible and insoluble matter in grams
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.4 Iron. Weigh 1.000 g of the crushed specimen (see 4.2.4.1) and dissolve in a solution of 3 ml of hydrochloric acid diluted with water to make 200 ml. Precipitate the copper with hydrogen sulfide and continue adding hydrogen sulfide until no further precipitation is observed. Filter and then boil a 10-ml aliquot of the filtrate to remove the excess hydrogen sulfide, add 3 ml of hydrochloric acid, and dilute to 100 ml with water. To 50 ml of this solution, add 0.1 g of ammonium persulfate crystals and 3 ml of 30-percent ammonium thiocyanate solution. Any red color produced shall be no darker than that produced by 0.05 milligram of iron (Fe) in an equal volume of solution containing the quantities of reagents used in the specimen test.

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4.2.4.5 Particle size. Determine particle size in accordance with Fed. Test Method Std. No. 536/2101. The 3/4-inch (19 mm) sieve used in this test shall conform to ASTM C136 and the 2.38 mm (No. 8) sieve shall conform to ASTM E11.

4.2.4.6 Appearance. Visually examine the specimen for color, form, and evidence of hard caking.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. Cupric sulfate shall be unit packed level A in a 2.27-kilogram (kg), 11.34-kg or 45.36-kg quantity as specified (see 6.2).

5.1.1.1 2.27-Kg quantity. A quantity of 2.27 (+0.02 or -0) kg of cupric sulfate shall be unit packed in a metal can or plastic bottle or jar. The metal can shall conform to PPP-C-186 with a wide mouth opening and shall be formed from tinplate having a minimum coating weight of commercial designation No. 100. When specified (see 6.2) the can shall be furnished with a polyethylene bag liner conforming to MIL-B-117, type III, class B, style optional and closure method optional. The exterior of all cans shall be coated and the side seam striped. The plastic bottle or jar shall conform to PPP-C-186, group A, class 2, style 2 and 4, respectively. The liners for the screw caps for cans, bottles, and jars shall be resistant to cupric sulfate. Closures shall be secured to the bodies of their respective containers assuring prevention of exposure of the cupric sulfate to the atmosphere and preventing sifting of contents. An outer or inner seal A or B, respectively, of PPP-C-186 shall be required.

(a) Intermediate packing. Six 2.27-kg unit packs of cupric sulfate prepared as specified in 5.1.1.1 shall be intermediately packed in a snug-fitting fiberboard box conforming to grade W5c as specified in PPP-C-186. Plastic unit packs shall be protected in close-fitting full container height cells formed by half-slotted partitions of fiberboard. Any motion of contents within the box shall be prevented by inserting fiberboard pads as needed. Fiberboard partitions and pads shall be formed from the same material as the box. The box shall be closed as specified in PPP-B-636 for the closure of interior containers.

5.1.1.2 11.34-Kg quantity. A quantity of 11.34 (+0.10 or -0) kg of cupric sulfate shall be unit packed in a plastic pail or in a steel pail with a plastic liner. The plastic pail shall conform to Department of Transportation (DOT) Specification 35. The steel pail shall conform to

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the level A requirements of PPP-C-186 with bolted locking ring. Each pail shall be furnished with a bag liner conforming to MIL-B-117, type II, class B, style optional, and closure method optional. The pails shall be closed securely to prevent contents from leaking or being exposed to the atmosphere.

5.1.1.3 45.36-Kg quantity. A quantity of 45.36 (+0.5 or -0) kg of cupric sulfate shall be unit packed in a steel pail conforming to the level A requirements of PPP-C-186 with bolted locking ring. The pail shall be furnished with a bag liner conforming to MIL-B-117, type II, class B, style optional and closure method optional. The pail shall be closed securely to prevent contents from leaking or being exposed to the atmosphere.

5.1.2 Level B. Cupric sulfate shall be unit packed in a 2.27-kg, 11.34-kg, or 45.36-kg quantity as specified (see 6.2).

5.1.2.1 2.27-Kg quantity. A quantity of 2.27 kg of cupric sulfate shall be unit packed as specified in 5.1.1.1 except that the side seam striping of the can shall not be required.

(a) Intermediate packing. Six 2.27-kg unit packs of cupric sulfate prepared as specified in 5.1.2.1 shall be intermediately packed as specified in 5.1.1.1(a) except that the box shall conform to the level B requirements of PPP-C-186. The material for the partitions and pads shall meet the same requirements as the box.

5.1.2.2 11.34-Kg quantity. A quantity of 11.34 kg of cupric sulfate shall be unit packed as specified in 5.1.1.2 except that the bolted locking ring shall not be required for the closure of the steel pail.

5.1.2.3 45.36-Kg quantity. A quantity of 45.36 kg of cupric sulfate shall be unit packed as specified in 5.1.1.3 except that the steel pail shall conform to the level B requirements of PPP-C-186.

5.1.3 Commercial (for civil agencies). Cupric sulfate shall be unit packed in the quantity specified (see 6.2), in accordance with normal commercial practice. The complete pack shall be designed to protect the cupric sulfate against damage during shipment, handling, and storage.

5.1.4 Commercial (for military activities). Cupric sulfate shall be unit packed in the quantity specified (see 6.2) in accordance with MIL-STD-1188.

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5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 2.27-Kg quantities. Four intermediate packs of cupric sulfate prepared as specified in 5.1.1.1(a) shall be packed level A, upright, in a close-fitting wood box for a type 2 load. The box shall be wirebound wood, nailed wood, or cleated-plywood. The wirebound wood box shall conform to PPP-B-585, class 3, style optional with wood parts preserved as specified for the grade A box of PPP-B-621. The nailed wood box shall conform to PPP-B-621, class 2, style 4, grade A. The cleated-plywood box shall conform to PPP-B-601, overseas type, style A, B, or I, grade A. Motion of contents within the box shall be prevented by inserting fiberboard pads as needed. Pads shall conform to PPP-F-320, minimum grade W5c. The box shall be closed and reinforced as specified in the applicable box specification.

5.2.1.2 11.34-Kg quantity. No further protection is required to ship unit packs of the 11.34-kg quantity prepared as specified in 5.1.1.2.

5.2.1.3 45.36-Kg quantity. No further protection is required to ship unit packs of the 45.36-kg quantity prepared as specified in 5.1.1.3.

5.2.2 Level B.

5.2.2.1 2.27-Kg quantity.

(a) Civil agencies. Six unit packs of the 2.27-kg quantity of cupric sulfate prepared as specified in 5.1.2.1 shall be packed upright in a close-fitting fiberboard box conforming to the level B requirements of PPP-C-186. Each unit pack shall be inserted in a close-fitting cell formed by interlocking half-slotted partitions of full box height. Full box-sized pads shall be inserted above and beneath the load. Additional pads shall be inserted in the box as needed to prevent motion of contents. Partitions and pads shall be formed from the same fiberboard as the box. The box shall be closed as specified in the box specification for class domestic boxes to be used as shipping containers.

(b) Military activities. Four intermediate packs of cupric sulfate prepared as specified in 5.1.2.1(a) shall be packed as specified in 5.2.1.1, except that the box shall be class domestic.

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5.2.2.2 11.34-Kg quantity. No further protection is required to ship unit packs of the 11.34-kg quantity prepared as specified in 5.1.2.2.

5.2.2.3 45.36-Kg quantity. No further protection is required to ship unit packs of the 45.36-kg quantity prepared as specified in 5.1.2.3.

5.2.3 Commercial (for civil agencies). Cupric sulfate unit packed as specified in 5.1.3 shall be packed in fiberboard boxes to insure delivery at destination, provide for redistribution by the initial receiving activity, and be acceptable by common carrier under National Motor Freight Classification and Uniform Freight Classification.

5.2.4 Commercial (for military activities). Cupric sulfate unit packed as specified in 5.1.4 shall be packed in accordance with MIL-STD-1188.

5.3 Unitization. Unitization shall be by palletization using MIL-STD-147 load type I for intermediate packs of the 2.27-kg quantity unit pack, load type III with extra blocking to stabilize the load for unit packs of the 11.34-kg quantity, and load type III for unit packs of the 45.36-kg quantity.

5.4 Marking. In addition to the precautionary marking specified in 5.4.1 for unit packs, all containers shall be marked to show the lot number and date of manufacture of the cupric sulfate. Marking for civil agencies shall be in accordance with Fed. Std. No. 123. For military activities, levels A and B unit packs, intermediate packs, packs, and pallet loads shall be marked in accordance with MIL-STD-129; commercial unit packs and packs shall be marked in accordance with MIL-STD-1188.

5.4.1 Precautionary marking. Each unit pack shall be marked durably and legibly as follows:

WARNING! Harmful if swallowed.
Wash thoroughly after handling.

FIRST AID: If swallowed, induce vomiting by sticking finger down throat or by giving soapy or strong salty water to drink. Repeat until vomit is clear. Call a physician. Never give anything by mouth to an unconscious person.

6. NOTES

6.1 Intended use. Cupric sulfate is intended for use in electroplating, in batteries, in manufacturing germicides or disinfectants, and in compounding insecticides.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Level of preservation required (see 5.1)
(c) Unit quantity required (see 5.1.1, 5.1.2, 5.1.3, and 5.1.4)
(d) If polyethylene bag is required for level A unit packing of 2.27-kg quantity in metal can (see 5.1.1.1)
(e) Level of packing required (see 5.2)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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